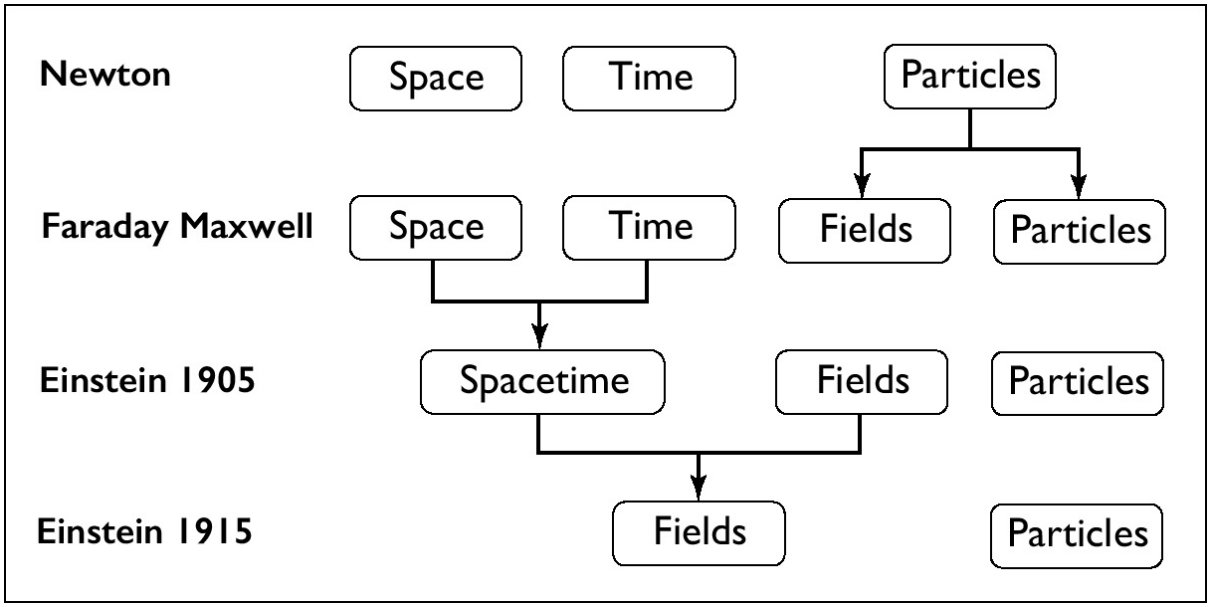



**Figure 3.5** What is the world made of?

The world is not made up of space + particles + electromagnetic field + gravitational field. The world is made up of particles + fields, and nothing else; there is no need to add space as an extra ingredient. Newton's space *is* the gravitational field. Or vice versa, which amounts to saying the same thing: the gravitational field is space (figure 3.5).

But, unlike Newton's space, which is flat and fixed, the gravitational field, by virtue of being a field, is something which moves and undulates, subject to equations – like Maxwell's field, like Faraday's lines.

It is a momentous simplification of the world. Space is no longer different from matter. It is one of the 'material' components of the world, akin to the electromagnetic field. It is a real entity which undulates, fluctuates, bends and contorts.

We are not contained within an invisible, rigid scaffolding: we are immersed in a gigantic, flexible mollusc (the metaphor is Einstein's). The Sun bends space around itself, and the Earth does not circle around it drawn by a mysterious distant force but runs straight in a space that inclines. It's like a bead which rolls in a funnel: there are no mysterious forces generated by the centre of the funnel, it is the curved nature of the funnel wall which guides the rotation of the bead. Planets circle around the Sun, and things fall, because space around them is curved (figure 3.6).